

(a) Explain the relationship between the gene mutation that causes sickle cell anaemia and the reduced oxygen-carrying capacity of red blood cells.

[5]

One risk associated with a blood transfusion is a condition known as transfusion-associated circulatory overload (TACO). TACO is caused by a large increase in blood volume over a short period of time. This increase in blood volume can be harmful.

..... [1]

Suggest how an increase in blood pressure in the pulmonary capillaries can cause fluid to enter the alveoli.

..... [1]

(c) People with kidney disease may be at a higher risk of TACO following a blood transfusion.

A study carried out in 2019 investigated whether there is a link between kidney disease and TACO. This study included data from:

- 200 people who developed TACO after a blood transfusion
- 405 people who did **not** develop TACO after a blood transfusion.

The people in the study were put into one of four categories:

- people with acute (short-term) kidney injury (AKI)
- people with chronic (long-term) kidney disease (CKD) who do **not** require dialysis
- people with severe chronic kidney disease who do require dialysis (CKD + D)
- people who do **not** have kidney disease.

In dialysis, a machine is used to carry out the function of the kidneys.

The results of this study are shown in Fig. 3.1.

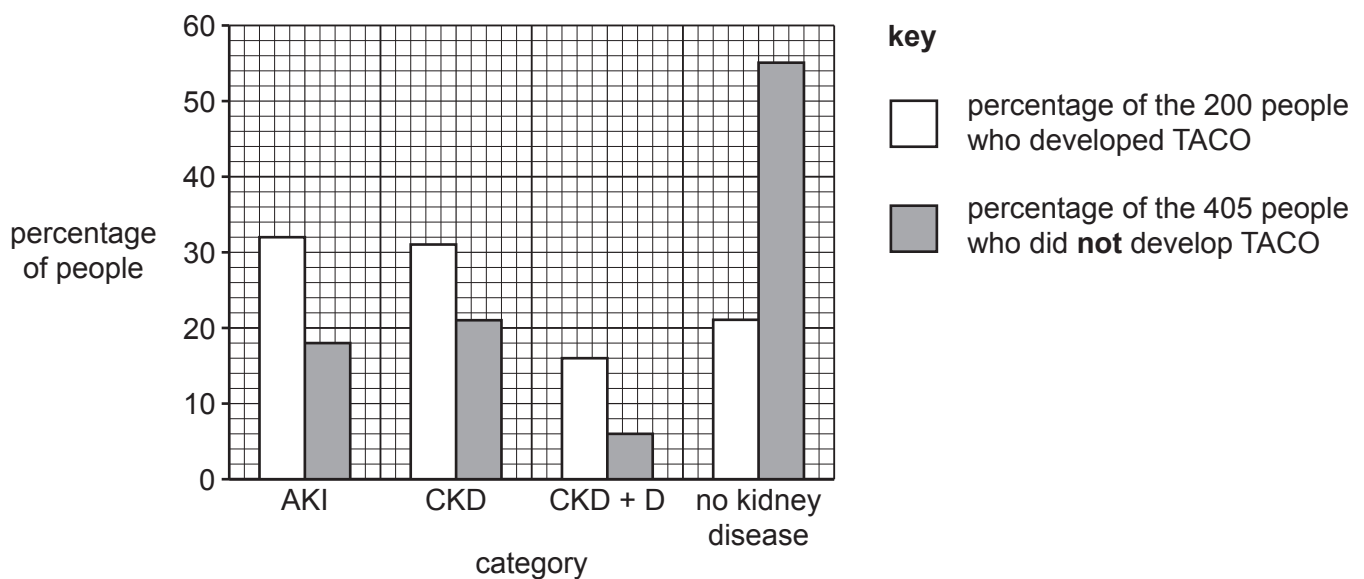


Fig. 3.1

With reference to Fig. 3.1, describe the effect of having kidney disease on the risk of developing TACO.

Suggest reasons for this effect.

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

- (d) Patients who are at high risk of developing TACO following a blood transfusion can be given a type of drug called a loop diuretic.

In normal urine production, 99% of sodium ions in the glomerular filtrate are reabsorbed and 1% are excreted. In urine production of people who take loop diuretics, 80% of sodium ions in the glomerular filtrate are reabsorbed and 20% are excreted.

Explain how loop diuretics affect water reabsorption in the kidneys **and** suggest why this reduces the risk of developing TACO following a blood transfusion.

.....

.....

.....

.....

.....

.....

..... [3]

[Total: 14]